

Dated: _____

Initial value

$$(ii) \quad \frac{dy}{dx} + \frac{e^x}{(e^x+1)} y = 3e^x (e^x+1) \quad y(0)=4$$

= Solution:

$$I.F = e^{\int \frac{e^x}{e^x+1} \cdot dx} = \ln(e^x+1)$$

$$I.F = e^x + 1$$

$$= \frac{dy}{dx} + e^x y = 3e^x (e^x+1)^2$$

Multiply I.F on B.S

$$(e^x+1) \left(\frac{dy}{dx} \right) + e^x y = 3e^x$$

$$= (e^x+1) \left(\frac{dy}{dx} \right) + (e^x+1) \left(\frac{e^x}{e^x+1} y \right) = (e^x+1) (3e^x (e^x+1)^2)$$

$$= (e^x+1) \frac{dy}{dx} + e^x y = 3e^x (e^x+1)^2 \rightarrow (1)$$

Product Rule:

$$\frac{d}{dx} (u \cdot v) = u' \cdot v + u \cdot v'$$

$$\Rightarrow u = y, v = e^x + 1 \quad \text{and} \quad \Rightarrow u' = \frac{dy}{dx}, v' = e^x$$

$$= \frac{d}{dx} [y(e^x+1)] = \frac{dy}{dx} (e^x+1) + y(e^x)$$

$$= \frac{d}{dx} [y(e^x+1)] = 3e^x (e^x+1)^2$$

Integrating on B.S w.r.t (x)

$$= \int \frac{d}{dx} [y(e^x+1)] \cdot dx = \int 3e^x (e^x+1)^2 \cdot dx$$

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$$= y(e^x + 1) = \int 3e^x (e^x + 1)^2 \cdot dx$$

$$= y(e^x + 1) = (e^x + 1)^3 + C \rightarrow \text{Chain rule}$$

$$\therefore y = \frac{(e^x + 1)^3 + C}{(e^x + 1)}$$

$$= y = \frac{(e^x + 1)^3}{(e^x + 1)} + \frac{C}{(e^x + 1)}$$

$$= y = (e^x + 1)^2 + \frac{C}{(e^x + 1)} \rightarrow (2)$$

Apply the condition $y(0) = 4$

$$= 4 = (e^0 + 1)^2 + \frac{C}{(e^0 + 1)}$$

$$= 4 = (1 + 1)^2 + \frac{C}{(1 + 1)}$$

$$= \cancel{4} = \cancel{4} + \frac{C}{2}$$

$$= \frac{C}{2} = 0$$

$$= C = 0$$

Put $(C = 0)$ in (2)

$$= y = (e^x + 1)^2 + 0 / (e^x + 1)$$

$$= y = (e^x + 1)^2$$

Ans

Dated: _____

(iii) $\frac{dy}{dx} = \frac{x(2\log x + 1)}{\sin y + y \cos y}$

Solve for DE

= Solution:

By Cross

$$= (\sin y + y \cos y) dy = x(2\log x + 1) dx$$

Int. on B.S

$$= \int (\sin y + y \cos y) dy = \int x(2\log x + 1) dx$$

L.H.S

$$= \int \sin y dy = -\cos y$$

$$= \int y \cos y dy = y \sin y - \int \sin y dy = y \sin y + \cos y$$

$$= -\cos y + y \sin y + C$$

R.H.S

$$\int x(2\log x + 1) dx$$

Int. by Parts:

$$= \int x(2\log x) dx = x^2 \log x - \int x \cdot \frac{2}{x} \cdot x dx = x^2 \log x - x^2 + C$$

$$= x^2 \log x - x^2 + C$$

$$= \frac{x^2}{2} (2\log x + 1) + C$$

Write the whole eq:

$$y \sin y - \cos y = x^2 \log x - x^2 + C$$

Ans

Dated:

(iv)

$$\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$$

Solve for DE

= Solution:

Let,

$$x = x+h, \quad y = y+k$$

So,

$$\frac{dy}{dx} = \frac{dy}{dx}$$

$$= \frac{dy}{dx} = \frac{x+2y+h+2k-3}{2x+y+2h+k-3} = \frac{x+2y}{2x+y}$$

$$\begin{aligned} h+2k-3 &= 0 \\ 2h+k-3 &= 0 \end{aligned} \quad \left(\begin{array}{l} h=1 \\ k=1 \end{array} \right)$$

As eq. is homogeneous and first order

Put $y = VX$

$$= V + X \frac{dV}{dx} = \frac{X+2VX}{2X+VX} = \frac{1+2V}{2+V}$$

$$= X \frac{dV}{dx} = \frac{1+2V - (2+V)V}{2+V} = \frac{1-V^2}{2+V}$$

Separate the var. and integrate

$$= \int \frac{2+V}{1-V^2} dV = \int \frac{dx}{x} +$$

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$$= \int \frac{2}{1-v^2} dv + \frac{1}{-2} \int \frac{-2v}{1-v^2} dv = \int \frac{dx}{x}$$

$$= \cancel{2} \left(\frac{1}{\cancel{2}} \ln \left| \frac{1+v}{1-v} \right| \right) - \frac{1}{2} \ln(1-v^2) = \ln x + C$$

$$= \ln \left| \frac{1+v}{1-v} \right| = -\ln \sqrt{1-v^2} = \ln x + C$$

$$\text{As, } \log a^n = n \log a$$

Ans

Dated: _____

Q#1) Obtain DE :

$$x^2 + y^2 + 2gx + 2fy = 0$$

= Solution:

Diff. w.r.t (x)

$$= \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) + \frac{d}{dx}(2gx) + \frac{d}{dx}(2fy) = \frac{d}{dx}(0)$$

$$= 2x + 2y \frac{dy}{dx} + 2g + 2f \frac{dy}{dx} = 0$$

Divide by 2:

$$= x + y \frac{dy}{dx} + g + f \frac{dy}{dx} = 0$$

$$= g = -x - y \frac{dy}{dx} - f \frac{dy}{dx}$$

Diff again w.r.t (x)

$$= \frac{d}{dx}(x) + \frac{d}{dx}\left(y \frac{dy}{dx}\right) + \frac{d}{dx}(g) + \frac{d}{dx}\left(f \frac{dy}{dx}\right)$$

$$= 1 + \left(y \cdot \frac{d^2 y}{dx^2} + \frac{dy}{dx} \cdot \frac{dy}{dx}\right) + 0 + \left(f \frac{d^2 y}{dx^2}\right)$$

$$= \left[1 + y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + f \cdot \frac{d^2 y}{dx^2} \right] = 0$$

↳ eq (1)

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$$= f \frac{d^2 y}{dx^2} = -1 - y \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx} \right)^2$$

$$= f = -1 - y \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx} \right)^2 \bigg/ \frac{d^2 y}{dx^2}$$

$$= f = -y - \frac{1 + (dy/dx)^2}{d^2 y / dx^2}$$

Diff. third time to eliminate constant (f)

$$= d/dx \left(1 + y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 + f \frac{d^2 y}{dx^2} \right) = 0$$

$$= 0 + \left(y \frac{d^3 y}{dx^3} + \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} \right) + \left(2 \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} \right) + \left(f \frac{d^3 y}{dx^3} \right) = 0$$

$$= y \cdot \frac{d^3 y}{dx^3} + 3 \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} + f \cdot \frac{d^3 y}{dx^3} = 0$$

$$\swarrow \text{eq (2)} \quad = (y + f) \cdot \frac{d^3 y}{dx^3} + 3 \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} = 0$$

From eq (1)

$$y + f = - \frac{1 + (dy/dx)^2}{d^2 y / dx^2}$$

Put into eq (2)

Obtain DE
MID

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$$\Rightarrow \left[- \frac{1 + (dy/dx)^2}{d^2y/dx^2} \right] \frac{d^3y}{dx^3} + 3 \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} = 0$$

multiply $- d^2y/dx^2$ on B.S

$$(1 + (dy/dx)^2) \cdot \frac{d^3y}{dx^3} - 3 \frac{dy}{dx} \left(\frac{d^2y}{dx^2} \right)^2 = 0$$

Ans

Initial value

Dated:

(iv) $(x^2 + 3y^2) dx - 2xy dy = 0$ $y(2) = 6$
= Solution:

$$= (x^2 + 3y^2) dx = 2xy dy$$

$$\left[\frac{dy}{dx} = \frac{x^2 + 3y^2}{2xy} \right]$$

$$\frac{dy}{dx} = \frac{x^2 + 3y^2}{2xy}$$

Let $y = vx$

$$\begin{aligned} \frac{v + x \frac{dv}{dx}}{dx} &= \frac{x^2 + 3(vx)^2}{2x(vx)} \rightarrow (1) \\ &= \frac{x^2(1 + 3v^2)}{2vx^2} = \frac{1 + 3v^2}{2v} \\ \Rightarrow x \frac{dv}{dx} &= \frac{1 + 3v^2}{2v} - v = \frac{1 + 3v^2 - 2v^2}{2v} \end{aligned}$$

$$x \frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$\frac{2v}{1 + v^2} \cdot dv = \frac{1}{x} \cdot dx$$

Integrating B.s

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$$= \int \frac{2v}{1+v^2} \cdot dv = \int \frac{1}{x} \cdot dx$$

$$= \ln |1+v^2| = \ln |x| + C$$

$$= \cancel{\ln |1+v^2|} = \cancel{\ln |x|} + C$$

$$= 1+v^2 = Cx$$

Substitute back:

$$v = y/x$$

$$1 + \left(y/x\right)^2 = Cx$$

$$1 + y^2/x^2 = Cx$$

$$x^2 + y^2/x^2 = Cx$$

$$x^2 + y^2 = Cx^3 \rightarrow \text{eq (2)}$$

$$\text{Put } (y(2)=6) \rightarrow (2)$$

$$2^2 + 6^2 = C(2)^3$$

$$4 + 36 = C(8)$$

$$C(8) = 40$$

$$C = 40/8$$

$$C = 5$$

$$\Rightarrow x^2 + y^2 = 5x^3$$

Ans

Dated:

Q#02) Solve the DE:

$$(a) \quad \frac{dy}{dx} = \frac{3xy + y^2}{3x^2}$$

= Solution:

$$= 3x^2 dy = (3xy + y^2) \cdot dx$$

$$= \frac{dy}{dx} = \frac{3xy + y^2}{3x^2}$$

Put $y = vx$

$$\text{So, } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$= v + x \frac{dv}{dx} = \frac{3vx^2 + v^2x^2}{3x^2}$$

$$= v + x \frac{dv}{dx} = \frac{3v + v^2}{3}$$

$$= x \frac{dv}{dx} = \frac{v^2}{3}$$

$$= 3 \int \frac{1}{v^2} \cdot dv = \int \frac{1}{x} \cdot dx$$

Integrating on B.S

$$= 3 \int \frac{1}{v^2} \cdot dv = \int \frac{1}{x} \cdot dx$$

$$= -3 \times \frac{1}{v} = \log |x| + C$$

$$= -3/v = \ln |x| + C$$

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Put $v = y/x$ back,

$$= \frac{-3x}{y} = \ln|x| + C$$

Ans
—.

Dated: _____

(Solve for DE)

$$(b) \quad \frac{dy}{dx} = \frac{y+x-2}{y-x-4}$$

= Solution:

Standard form

$$\frac{dy}{dx} = \frac{A_1x + B_1y + C_1}{A_2x + B_2y + C_2}$$

Separable

$$1) \quad \frac{A_1}{A_2} = \frac{B_1}{B_2}$$

✓ 2)

Reduce to homogeneous

$$\frac{A_1}{A_2} \neq \frac{B_1}{B_2}$$

So,

$$x = X+h, \quad y = Y+k \quad \text{and} \quad \frac{dy}{dx} = \frac{dY}{dX}$$

$$= \frac{dY}{dX} = \frac{Y+k+X+h-2}{Y+k-(X+h)-4}$$

$$= \frac{dY}{dX} = \frac{Y+X+(h+k-2)}{Y-X+(k-h-4)}$$

$$= \frac{dY}{dX} = \frac{Y+X}{Y-X}$$

$$= Y = VX$$

$$= \frac{dY}{dX} = V + X \frac{dV}{dX}$$

$$= V + X \frac{dV}{dX} = \frac{VX+X}{VX-X}$$

$$= V + X \frac{dV}{dX} = \frac{V+1}{V-1}$$

$$\left\{ \begin{array}{l} h+k-2=0 \\ -h+k-4=0 \\ \hline 2k-6=0 \\ \boxed{k=3} \end{array} \right\}$$

for h,

$$h+3-2=0$$

$$h+1=0$$

$$\boxed{h=-1}$$

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$$= x \frac{dv}{dx} = \frac{v+1}{v-1} - v \Rightarrow \frac{x dv}{dx} = \frac{v+1-v^2+v}{v-1}$$

$$= \frac{x dv}{dx} = \frac{-(v^2-2v-1)}{v-1} \Rightarrow \frac{v-1}{v^2-2v-1} dv = -\frac{dx}{x}$$

Int. on BS

$$= \int \frac{v-1}{v^2-2v-1} dv = - \int \frac{dx}{x}$$

$$\text{Let, } v^2-2v-1=t \Rightarrow (2v-2)dv = dt \\ = (v-1)dv = \frac{1}{2}dt$$

$$= \frac{1}{2} \int \frac{dt}{t} = -\ln x + \ln C \Rightarrow \frac{1}{2} \ln(v^2-2v-1) + \ln x = \ln C$$

$$= \ln(v^2-2v-1)^{\frac{1}{2}} + \ln x = \ln C \quad (\text{Put } v = y/x)$$

$$= \ln \sqrt{\frac{y^2}{x^2} - \frac{2y}{x} - 1} + \ln x = \ln C$$

$$= \ln \sqrt{\frac{y^2 - 2xy - x^2}{x^2}} + \ln x = \ln C$$

$$= \ln \frac{1}{x} \sqrt{y^2 - 2xy - x^2} + \ln x = \ln C$$

$$= \ln \sqrt{y^2 - 2xy - x^2} = \ln C \Rightarrow \ln(y^2 - 2xy - x^2)^{\frac{1}{2}} = \ln C$$

$$= \frac{1}{2} \ln(y^2 - 2xy - x^2) = \ln C$$

$$= \ln(y^2 - 2xy - x^2) = 2 \ln C \Rightarrow \ln(y^2 - 2xy - x^2) = \ln C$$

Let $C = a$, Put $h = -1$, $k = 3$

$$= y^2 - 2xy - x^2 = a \quad (y = y+k, x = x+h)$$

$$= (y-3)^2 - 2(x+1)(y-3) - (x+1)^2 = a$$

Ans

Dated: /

Initial value :

(v) $(x^2 + 3y^2)dx - 2xydy = 0$, $y(0) = 4$
= Solution:

$$= \frac{dy}{dx} = \frac{x^2 + 3y^2}{2xy} \rightarrow \text{eq, (1)}$$

$$\text{Let } y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$= v + x \frac{dv}{dx} = \frac{1 + 3v^2}{2v}$$

$$= x \frac{dv}{dx} = \frac{1 + 3v^2 - 2v^2}{2v}$$

$$= x \frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$= \frac{2v}{1 + v^2} dv = \frac{dx}{x} \rightarrow \text{eq, (2)}$$

Int. on B.S

$$\int \frac{2v}{1 + v^2} dv = \int \frac{dx}{x}$$

$$\text{Put } \rightarrow 1 + v^2 = t$$

$$2v dv = dt$$

$$\int \frac{dt}{t} = \int \frac{dx}{x}$$

$$= \ln |t| = \ln |x| + \ln |C_1|$$

$$= \ln \left| \frac{t}{x} \right| = \ln |C_1|$$

$$= \frac{t}{x} = \pm C_1$$

$$= \frac{1 + v^2}{x} = \pm C_1$$

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$$= \frac{1+y^2}{x^2} = \pm C$$

$$\left(\frac{v = y/x}{x} \right)$$

$$= x^2 + y^2 = C x^3$$

$$= C = x^2 + y^2 / x^3$$

Ans

Dated:

Q. 01) Obtain DE of circle of "r" = "a" with centre at (h, k).
radius

= Solution:

$$= (x-h)^2 + (y-k)^2 = a^2$$

Diff. w.r.t (x)

$$= 2(x-h) + 2(y-k) \frac{dy}{dx} = 0$$

Divide by (2)

$$= (x-h) + (y-k) \frac{dy}{dx} \rightarrow (1)$$

Diff. w.r.t (x) again

$$= 1 + \left(\frac{dy}{dx} \right) \frac{dy}{dx} + (y-k) \frac{d^2y}{dx^2} = 0$$

$$= 1 + (y')^2 + (y-k)y'' = 0$$

$$= y-k = - \frac{1 + (y')^2}{y''} \rightarrow (2)$$

From (1) and (2)

$$= \left(- (y-k) \frac{dy}{dx} \right)^2 + (y-k)^2 = a^2,$$

$$= (y-k)^2 \left(\frac{dy}{dx} \right)^2 + (y-k)^2 = a^2.$$

Substitute eq (2) into above eq:

$$= \left(- \frac{1 + (y')^2}{y''} \right)^2 (y')^2 + \left(- \frac{1 + (y')^2}{y''} \right)^2 = a^2$$

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$$= \left(1 + \frac{(y')^2}{y''}\right)^2 \left((y')^2 + 1\right) = a^2$$

$$= \boxed{(1 + (y')^2)^3 = a^2 (y'')^2}$$

Ans
— .

Dated:

R # 02) Solve DE :

$$a) \quad \frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$$

= Solution: (Already solved)

$$b) \tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$$

$$(b) \quad \tan y \cdot \frac{dy}{dx} + \tan x = \cos y \cos^2 x$$

= Solution:

$$\tan y \cdot \frac{dy}{dx} + \tan x = \cos y \cos^2 x \rightarrow (1)$$

Multiply eq (1) by $\sec y$

$$= \sec y \tan y \cdot \frac{dy}{dx} + \sec y \tan x = \sec y \cos y \cos^2 x$$

As, $\sec y \cos y = 1$, then:

$$= \sec y \tan y \cdot \frac{dy}{dx} + \sec y \tan x = \cos^2 x$$

Put $\sec y = t$

$$= \frac{dt}{dx} = \sec y \tan y \frac{dy}{dx}$$

$$= \frac{dt}{dx} + \frac{t \tan x}{f(x)} = \frac{\cos^2 x}{g(x)} \rightarrow (2)$$

Find I.F

$$= I.F = e^{\int f(x) dx} = e^{\int \tan x dx} = e^{-\ln(\cos x)} = \sec x$$

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Multiply eq(2) by I.F

$$= \sec x \cdot \frac{dt}{dx} + \sec x \cdot \tan x t = \sec x \cdot \cos^2 x$$

$$= \sec x \frac{dt}{dx} + t \cdot \sec x \cdot \tan x = \sin^2 x$$

L.H.S can be expressed as der. of

$$\frac{d}{dx} (t \sec x) = \sin^2 x$$

Int. on B.S

$$= \int \frac{d}{dx} (t \sec x) dx = \int \sin^2 x dx$$

$$= t \sec x = \int \sin^2 x dx + C$$

$$\text{As, } \sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$= \int \sin^2 x dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

$$= t \sec x = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

Substitute $t = \sec y$.

$$= \sec y \sec x = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

$$= \sec y = \frac{\frac{x}{2} - \frac{\sin(2x)}{4}}{\sec x} + C$$

Ans